

TAMBOV, Russian Federation

WMO: 279470

Lat: 52.800N

Lon: 41.333E

Elev: 128

StdP: 99.79

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.2	-23.0	-28.7	0.3	-25.9	-25.7	0.4	-22.7	10.3	-2.5	9.3	-3.9	1.6	350	0.578

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	32.2	20.0	30.1	19.5	28.2	18.8	21.6	29.3	20.7	27.8	19.8	26.3	3.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	14.1	24.3	18.2	13.3	23.6	17.4	12.6	22.8	63.4	29.5	60.2	27.8	57.2	26.4	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
9.2	7.8	7.0	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-29.4	34.4	3.6	2.2	-32.0	35.9	-34.2	37.2	-36.2	38.5	-38.9	40.1
			-29.5	23.0	3.6	1.0	-32.1	23.7	-34.3	24.2	-36.3	24.8	-38.9	25.5
										</				

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	6.4	-8.1	-7.7	-2.2	7.7	14.9	18.1	20.6	19.0	13.3	6.4	-0.3	-5.5	(6)
(7)		DBStd	11.61	7.22	7.12	5.09	4.97	4.84	4.01	3.52	4.02	4.30	4.58	5.37	6.58	(7)
(8)		HDD10.0	2491	563	494	379	103	12	1	0	0	17	131	311	479	(8)
(9)		HDD18.3	4574	821	728	637	321	129	53	16	41	159	371	561	738	(9)
(10)		CDD10.0	1181	0	0	0	33	163	244	329	279	114	18	1	0	(10)
(11)		CDD18.3	222	0	0	0	1	21	46	87	61	7	0	0	0	(11)
(12)		CDH23.3	2348	0	0	0	9	237	478	867	668	88	0	0	0	(12)
(13)		CDH26.7	793	0	0	0	1	57	149	316	256	16	0	0	0	(13)
(14)	Wind	WSAvg	3.4	3.8	4.0	3.9	3.6	3.3	3.0	2.7	2.8	3.0	3.5	3.9	3.9	(14)
(15)	Precipitation	PrecAvg	528	37	33	29	29	45	70	55	48	50	49	43	40	(15)
(16)		PrecMax	756	68	70	62	98	146	155	132	117	130	104	110	84	(16)
(17)		PrecMin	340	17	6	3	6	12	27	9	8	2	17	2	9	(17)
(18)		PrecStd	110	13	17	14	20	32	31	31	27	33	23	29	19	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.6	3.5	13.0	24.7	30.7	32.8	35.7	36.5	28.7	20.7	12.3	7.4	(19)
(20)			MCWB	2.7	2.4	8.0	13.2	18.3	19.1	21.5	21.4	16.8	13.8	10.5	6.6	(20)
(21)		2%	DB	2.0	2.5	8.7	21.6	28.3	30.2	32.2	32.2	25.9	17.8	10.0	4.8	(21)
(22)			MCWB	1.2	1.6	5.2	12.3	17.2	19.5	20.6	19.9	16.1	12.7	8.8	3.9	(22)
(23)		5%	DB	1.3	1.7	6.1	19.1	26.1	28.3	30.0	29.3	23.3	15.2	8.2	2.9	(23)
(24)			MCWB	0.6	0.9	3.5	11.3	16.4	19.1	20.1	19.2	15.6	11.6	7.0	2.1	(24)
(25)		10%	DB	0.6	1.0	4.0	16.6	23.8	26.1	28.0	26.9	20.8	13.1	6.3	1.6	(25)
(26)			MCWB	0.0	0.3	2.2	10.1	15.4	18.1	19.3	18.3	14.5	10.5	5.3	0.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.9	2.8	8.2	14.5	19.8	22.1	22.8	22.5	18.8	15.9	10.8	6.6	(27)
(28)			MCD B	3.4	3.2	12.1	20.8	28.2	28.4	31.2	33.0	24.9	19.2	12.1	7.1	(28)
(29)		2%	WB	1.4	1.8	5.7	13.0	18.3	21.0	21.8	21.1	17.4	13.6	8.9	4.0	(29)
(30)			MCD B	1.8	2.2	8.3	20.0	25.9	27.6	29.7	28.9	22.8	16.2	9.9	4.6	(30)
(31)		5%	WB	0.7	1.1	3.6	11.9	17.1	19.9	20.9	20.1	16.3	12.1	7.2	2.2	(31)
(32)			MCD B	1.2	1.6	5.6	17.9	24.6	26.2	28.3	27.4	21.8	14.7	8.1	2.7	(32)
(33)		10%	WB	0.1	0.4	2.3	10.7	16.0	18.9	20.1	19.1	15.2	10.6	5.4	1.1	(33)
(34)			MCD B	0.7	1.0	3.8	15.9	22.5	24.7	26.5	25.3	20.0	12.9	6.2	1.5	(34)
(35)	Mean Daily Temperature Range	MDBR	5.8	6.6	7.3	10.3	11.5	10.6	11.1	11.7	10.3	7.3	4.9	5.1	(35)	
(36)		5% DB	MCD BR	3.7	4.4	9.6	14.2	14.3	13.9	14.0	15.4	14.4	10.9	6.6	3.8	(36)
(37)			MCWBR	3.7	4.1	6.4	6.9	6.0	5.6	5.2	5.8	6.5	6.8	5.5	3.6	(37)
(38)		5% WB	MCD BR	3.6	4.3	8.3	12.6	13.0	12.1	12.4	13.0	12.1	9.4	5.8	4.0	(38)
(39)			MCWBR	3.7	4.3	5.9	6.6	6.1	5.6	5.2	5.6	6.2	6.7	5.2	3.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.299	0.339	0.364	0.432	0.396	0.391	0.417	0.439	0.386	0.363	0.336	0.305	(40)	
(41)		taud	2.099	2.008	2.052	2.098	2.313	2.348	2.279	2.205	2.339	2.328	2.240	2.126	(41)	
(42)		Ebn at Noon	649	729	804	796	855	862	828	782	786	718	611	565	(42)	
(43)		Edh at Noon	73	110	130	142	120	118	124	127	98	81	66	61	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.87	1.87	3.38	4.28	5.75	5.89	5.85	4.90	3.24	1.63	0.71	0.48	(44)	
(45)		RadStd	0.20	0.25	0.50	0.35	0.49	0.62	0.48	0.43	0.40	0.25	0.16	0.15	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.71	N/A	N/A	N/A	N/A	N/A	N/A	-153	+128	N/A
(47)	Regional (8 neighbors)	+0.80	N/A	N/A	N/A	N/A	N/A	N/A	-177	+132	N/A

Nomenclature: See separate page